UNIT 2: SHAPES AND REGIONS

SESSION 1: BINARY SHAPE ANALYSIS

SLO 2: APPLY SHAPE DESCRIPTORS AND MORPHOLOGICAL TECHNIQUES TO COMPARE AND ANALYZE BINARY SHAPES FOR PATTERN RECOGNITION TASKS.

Activity 1: Shape Descriptor Analysis

Observe the following binary shapes.

Shape A

0 0 0 0 0
0 1 1 1 0
0 1 1 1 0
0 0 0 0 0

Shape B

0 0 0 0 0
0 1 1 0 0
0 1 1 0 0
0 1 1 0 0
0 0 0 0 0

Answer the following:

1. **Area of Shape A** (number of object pixels) = 6
2. **Area of Shape B** = 6
3. **Width of Shape A** = 3 pixels
4. **Height of Shape B** = 3 pixels
5. Which shape has a **larger aspect ratio**?
   Shape B has a larger aspect ratio.

6. Based on the shape descriptors, explain how these two shapes are **different**.

   Shape A is wider and shorter, with a width of 3 pixels and height of 2 pixels. Shape B is narrower and taller, with a width of 2 pixels and height of 3 pixels. Both shapes have the same area of 6 pixels, but their aspect ratios and overall shapes are different.

Activity 2: Morphological Operations

Consider the binary shape below.

0 0 0 0 0
0 1 1 0 0
0 1 1 0 0
0 0 0 0 0

Apply the following operations conceptually.

1. If **dilation** is applied, what will happen to the object?

☐ Object will expand
☐ Object will shrink
☐ Object disappears

Explain briefly:

Dilation adds pixels around the boundary of the object, making the object larger and thicker

2. If **erosion** is applied, what will happen to the object?

☐ Object will expand
☐ Object will shrink
☐ Object remains same
Explain briefly: Erosion removes pixels from the boundary of the object, making the object smaller and thinner.

3. Why are **morphological operations important before pattern recognition**?
4.

Morphological operations are important before pattern recognition because they help remove noise, fill small gaps or holes, and improve the shape of objects. This produces cleaner binary images and makes it easier to extract accurate features for recognition.

Reflection Question

How do **shape descriptors and morphological operations together help in object classification?**
Shape descriptors provide information such as area, width, height, and aspect ratio, which helps distinguish different objects. Morphological operations clean and improve the shapes before these features are extracted. Together, they provide reliable shape information that can be used to compare and classify objects accurately